Cooldown

> The recipe of record ends with a cooldown: over the last 40 epochs the pull and the anti weight both fall to a tenth. We branched each of eight runs at the start of the cooldown and trained one branch with it and one without. The two branches ended the same on every measurement we take, so the cooldown can go.

The cooldown was in the schedule from the first D2.2 runs, and nothing since has tested it. By epoch 360 the learning rate is a few percent of its peak, so little should still be moving, and a recipe without it would be simpler to describe. This report carries the *fixed-confirm*° runs at seeds 712 to 719 on from their stored epoch-64 states to epoch 360, and branches each there into two arms that differ only in the cooldown. Because the arms share everything up to epoch 360, we compare them seed by seed.

fixed-confirm. `τ_ℓ fixed` kept the anchor on all eight runs at fresh seeds with removal near 0.9 and no run over the spill criterion, and became the recipe of record.

The branches differ by a few thousandths or less on removal, spill°, the task score of every op, the loss, and where the op sits on e_1 (E1). Neither arm has a run over the spill criterion.

Spill. The side effect of an edit on the other ops…

Observations

- The two arms at epoch 400 (E1): without the cooldown, removal is higher by 0.001 on average (95% interval −0.001 to 0.004), and every other difference is as small or smaller.

Scope and method

Every run is the recipe of record, `τ_ℓ fixed`, at the seeds of the *fixed-confirm*° experiment. The `cooldown` arm runs it as it is. The `no cooldown` arm takes the end anneal out of both terms, so the pull and the anti weight hold their epoch-360 values to the end. The learning rate schedule is the same in both. The stems and both arms trained and were scored on L40S cards. The stored runs trained on L4s, so the cooldown arm is a close repeat of them rather than the same run.

fixed-confirm. `τ_ℓ fixed` kept the anchor on all eight runs at fresh seeds with removal near 0.9 and no run over the spill criterion, and became the recipe of record.

The measurements are those of the *fixed-confirm* experiment: removal and spill° under the full edit, net of the median over the other 63 axes of the run, spill under the edit at block 1 alone, the score per op, and α° by slice.

Spill. The side effect of an edit on the other ops…

Alignment. How far a state points along e₁…

The two arms at epoch 400 (E1)

The table compares the arms seed by seed. The main measure is removal, and the differences on it are a tenth of the seed range or less. Most of the paired intervals exclude zero, because two branches from one state stay close enough that a gap of a thousandth shows up. But no gap is large enough to matter for any choice we make with these numbers: the largest are the color embeddings leaning a little further onto e_1 without the cooldown, and α° at slices 1 and 2 a few thousandths higher. Neither arm has a run over the spill° criterion, in the full edit or at block 1 alone.

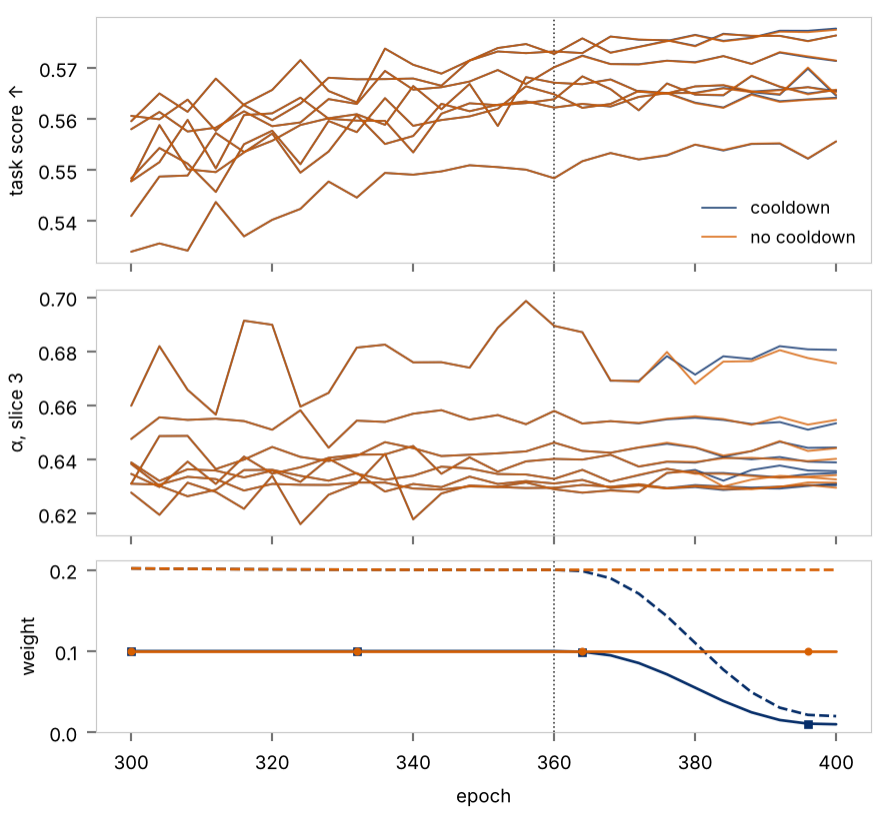
Alignment. How far a state points along e_1…

Spill. The side effect of an edit on the other ops…

measurement	cooldown	no cooldown	difference [95% interval]
removal ↑	0.879	0.880	0.001 [−0.001, 0.004]
spill ↓	−0.003	−0.003	0.000 [−0.000, 0.000]
spill, block 1 ↓	0.003	0.003	0.001 [0.000, 0.001]
validation loss ↓	2.1879	2.1879	−0.0000 [−0.0001, −0.0000]
embedding lean	−0.41	−0.42	−0.01 [−0.01, −0.00]
score, `mix` ↑	0.284	0.284	−0.000 [−0.000, 0.000]
score, `lighten` ↑	0.718	0.718	−0.000 [−0.000, −0.000]
score, `darken` ↑	0.710	0.710	−0.000 [−0.000, 0.000]
score, `difference` ↑	0.723	0.724	0.000 [0.000, 0.001]
score, `hue-hsv` ↑	0.588	0.588	0.000 [−0.000, 0.000]
score, `sat-hsv` ↑	0.481	0.481	−0.000 [−0.000, 0.000]
score, `value-hsv` ↑	0.487	0.487	0.000 [−0.000, 0.000]
α, embedding	0.02	0.02	−0.00 [−0.00, −0.00]
α, slice 1	0.31	0.32	0.00 [0.00, 0.01]
α, slice 2	0.55	0.56	0.00 [0.00, 0.01]
α, slice 3	0.64	0.64	−0.00 [−0.00, 0.00]
α, slice 4	0.49	0.49	0.00 [−0.00, 0.00]

***The two arms at epoch 400**, as seed means over eight runs, and the mean of the differences seed by seed (no cooldown less cooldown) with its 95% paired t interval. Removal and spill are net of the median over the other 63 axes of the run, and spill is the largest net drop on another op at any dose. The scores are expected exact match per op on the held-out contexts. α is at the example answers of the `difference` contexts, by slice.*

The figure shows why so little moves. By epoch 360 the task score is rising only slowly and α at slice 3 is about flat, and dropping both weights to a tenth does not change either line. The learning rate is a few percent of its peak by then, so the model has little room to respond to the terms in either arm.



***The last part of training, by arm.** One line per run. **Top:** expected exact match, a mean over ops, on the held-out contexts recorded along training. **Middle:** α at the example answers of the `difference` probe contexts at slice 3. **Bottom:** the pull weight (solid) and the anti weight at the embedding (dashed). The arms share every epoch before the dotted line.*

References

fixed-confirm

`τ_ℓ fixed` kept the anchor on all eight runs at fresh seeds with removal near 0.9 and no run over the spill criterion, and became the recipe of record.

paired-runs

Runs branched from one stored state stay alike seed by seed for well over a hundred epochs, so arms that share a state can be compared in pairs.